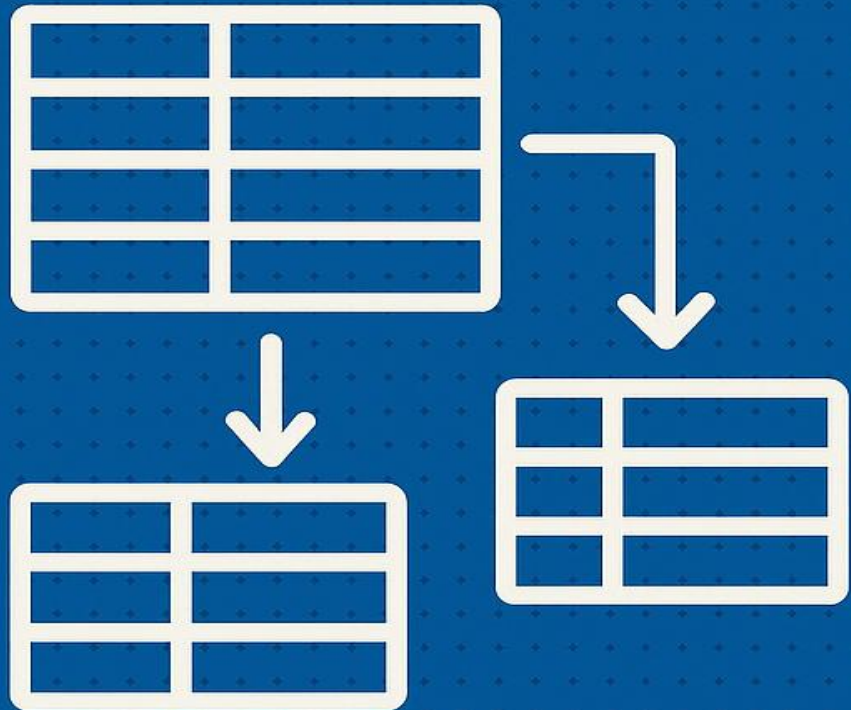
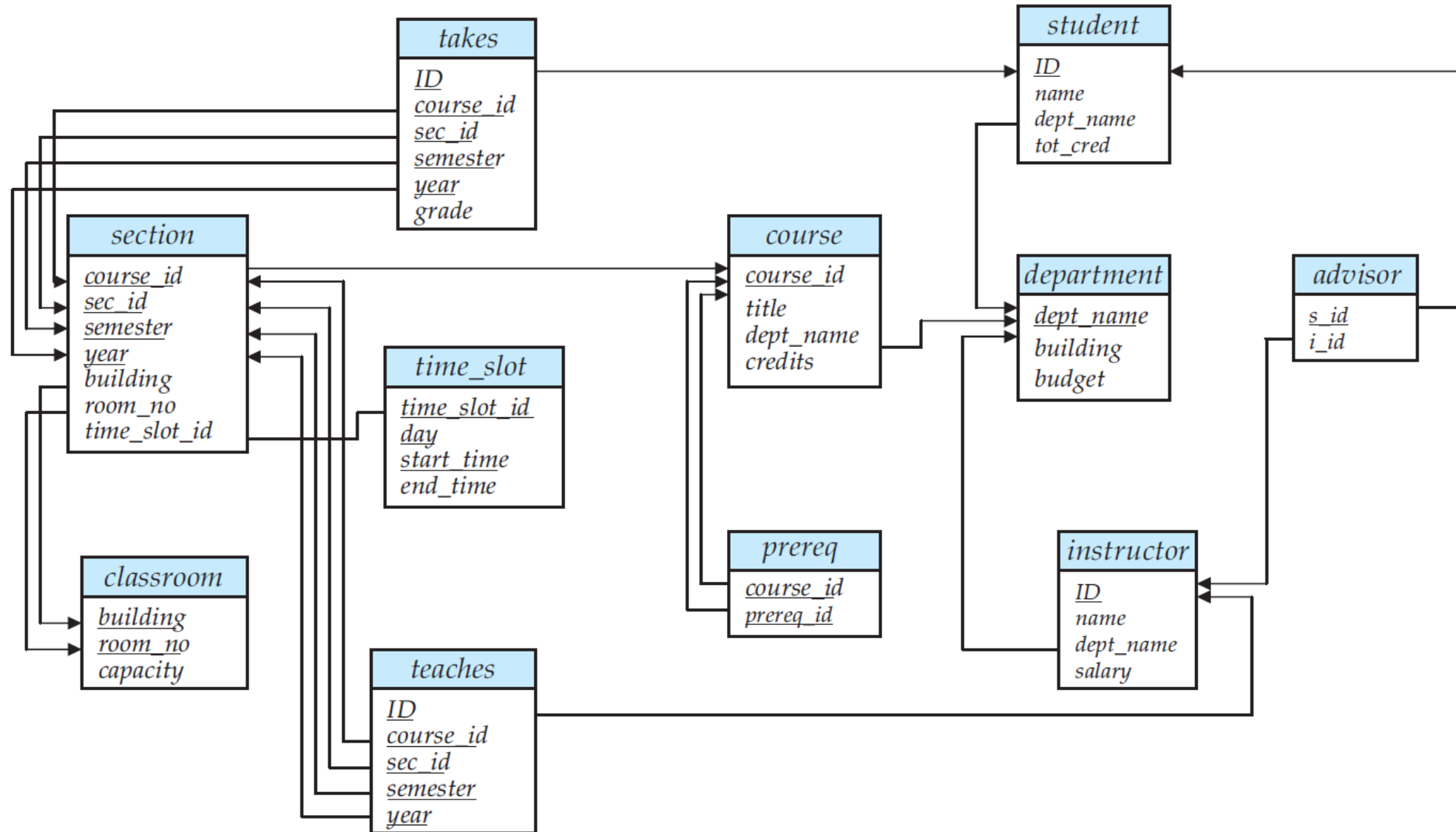


DATABASE NORMALIZATION



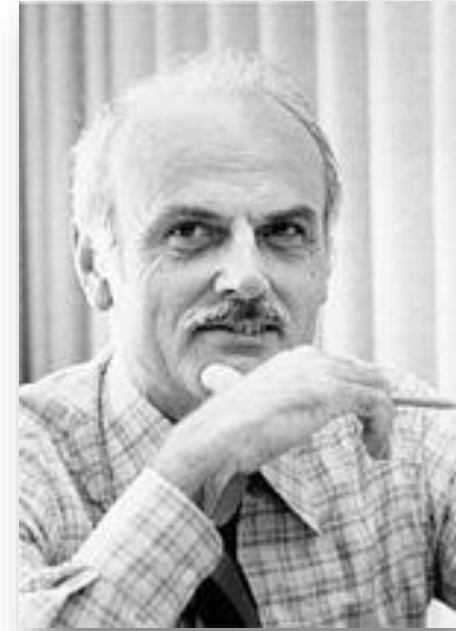
Schema Diagram

University Database



Database normalization

- The process of **structuring** a relational database in accordance with a series of so-called **normal forms**:
 - To reduce data **redundancy**
 - To improve **data integrity**.
- Normalization entails organizing the **attributes** and **relations** of a database
 - To ensure that their dependencies are properly enforced by database integrity constraints.
- Procedure:
 - A process of synthesis,
 - e.g. creating a new database design
 - Decomposition,
 - e.g. improving an existing database design.



Edgar Frank Codd
(1923~2003)

Combined Schemas?

- Consider a combination of *instructor* and *department* into *inst_dept*
 - No connection to relationship set *inst_dept*
 - Redundancy**

<i>ID</i>	<i>name</i>	<i>salary</i>	<i>dept_name</i>	<i>building</i>	<i>budget</i>
22222	Einstein	95000	Physics	Watson	70000
12121	Wu	90000	Finance	Painter	120000
32343	El Said	60000	History	Painter	50000
45565	Katz	75000	Comp. Sci.	Taylor	100000
98345	Kim	80000	Elec. Eng.	Taylor	85000
76766	Crick	72000	Biology	Watson	90000
10101	Srinivasan	65000	Comp. Sci.	Taylor	100000
58583	Califieri	62000	History	Painter	50000
83821	Brandt	92000	Comp. Sci.	Taylor	100000
15151	Mozart	40000	Music	Packard	80000
33456	Gold	87000	Physics	Watson	70000
76543	Singh	80000	Finance	Painter	120000

A Combined Schema Without Repetition

- Consider combining relations

sec_class(sec_id, building, room_number)

and

section(course_id, sec_id, semester, year)

- into one relation

*section(course_id, sec_id, semester, year,
building, room_number)*

- No repetition in this case

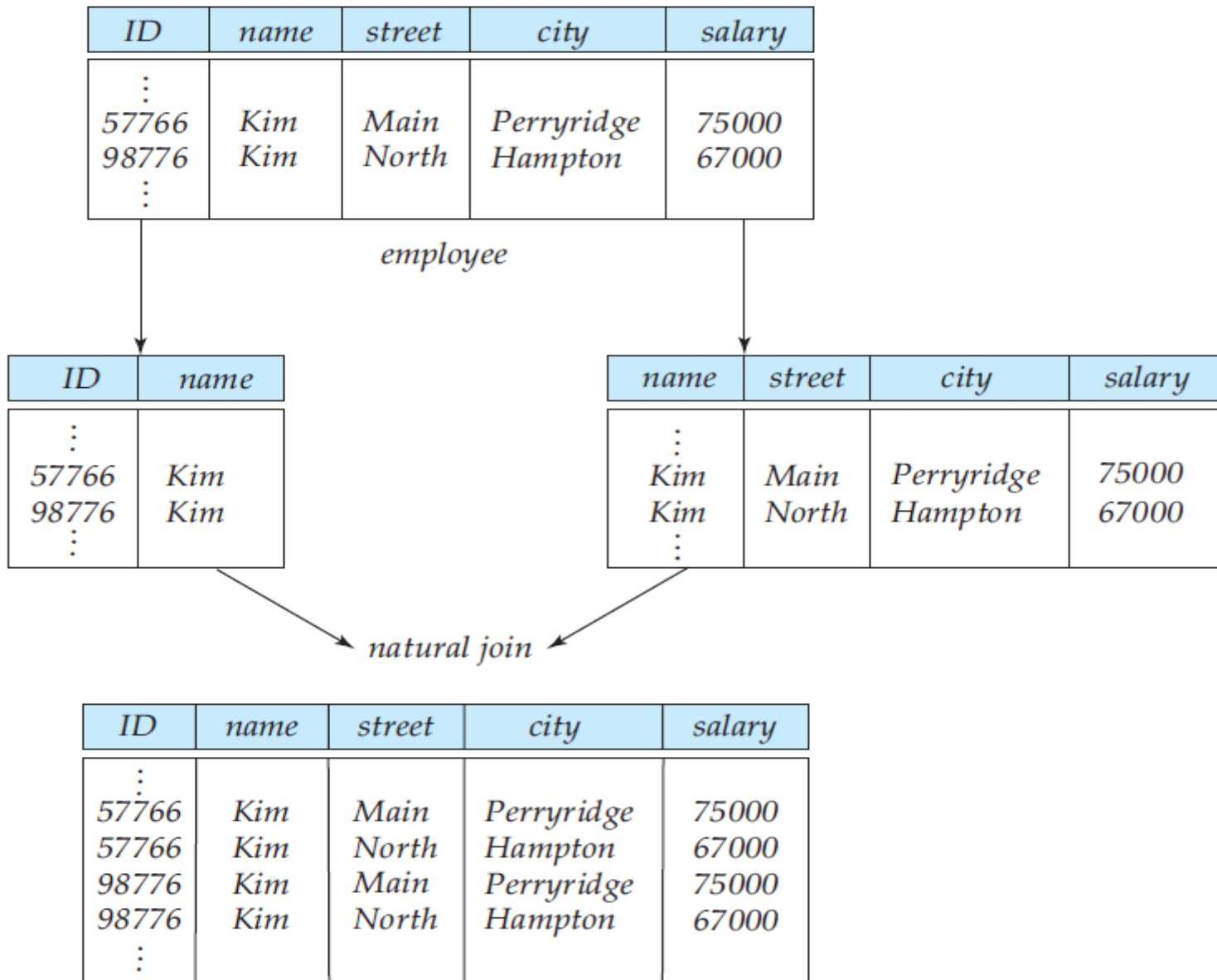
Functional Dependency

- **Functional dependency:**
 - A constraint that determines the relation of one attribute to another attribute.
 - To maintain the quality of data in the database
 - Denoted by $\rightarrow$

$ID \rightarrow name, street, city, salary$

<i>ID</i>	<i>name</i>	<i>street</i>	<i>city</i>	<i>salary</i>
⋮				
57766	Kim	Main	Perryridge	75000
98776	Kim	North	Hampton	67000
⋮				

A Lossy Decomposition



Lossless-Join Decomposition

- Decomposition of $R = (A, B, C)$ into $R_1 = (A, B)$ $R_2 = (B, C)$

A	B	C
α	1	A
β	2	B

r

A	B
α	1
β	2

$\Pi_{A,B}(r)$

B	C
1	A
2	B

$\Pi_{B,C}(r)$

$\Pi_A(r) \bowtie \Pi_B(r)$
Natural Join

A	B	C
α	1	A
β	2	B

Normalization Forms

- First Normal Form (**1NF**)
- Second Normal Form (**2NF**)
- Third Normal Form (**3NF**)
- Boyce-Codd Normal Form (**BCNF**)
- Forth Normal Form (**4NF**)
- Fifth Normal Form (**5NF**)
- Sixth Normal Form (**6NF**)

First Normal Form (1NF)

- **Conditions:**
 - Rule #1: **atomic values**
 - Rule #2: domains with the **same type**
 - Rule #3: **unique name** for each attribute
 - Rule #4: **order**
- **Advantage:**
 - Storage efficiency
 - No redundancy

- **Atomicity**

- How the elements of the domain are used.
- e.g. Strings would normally be considered indivisible
- Example: STIDs: *CS0012* or *EE1127*

1NF

- **Rule #1**

- Each column should contain **atomic values**
 - Entries like X,Y and W,X violate this rule.



column 1	column 2	column 3
A	X, Y	...
B	W, X	...
C	Y	...
D	Z	...

1NF

- **Rule #2**

- A column should contain values that are of the **same type**.
 - Don't inter-mix different types of values in any column.

ID	name	dept_name	tot_cred
210101	Park	CSE	8
210102	Lee	CSE	10
210103	Choi	CSE	12
Han	210104	CSE	8

- **Rule #3**

- Each column should have a **unique name**.
 - The same names leads to confusion at the time of data retrieval.



ID	name	name
210101	Park	Yongwoo
210102	Lee	Dohyeon
210103	Choi	Jihyeok
210104	Han	Jaeil

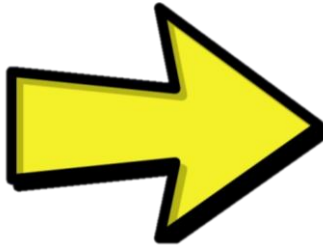
1NF

- **Rule #4**
 - **Order** in which data is saved doesn't matter.

ID	Last_name	First_name
210104	Park	Yongwoo
210103	Lee	Dohyeon
210102	Choi	Jihyeok
210101	Han	Jaeil

1NF

ID	name	course_title
210101	Kim	OS, CN
210103	Park	Java
210102	Lee	C,C++



ID	name	course_title
210101	Kim	OS
210101	Kim	CN
210103	Park	Java
210102	Lee	C
210102	Lee	C++

Second Normal Form (2NF)

- **Conditions:**
 - Be in **1NF**.
 - No **Partial Dependency**.

ID	name	dept_name	tot_cred
210101	Kim	CSE	10
210103	Kim	IT	12

- **Functional Dependency:** Every other column **depends** on **ID**.

Functional Dependencies

- R as a relation schema

$$\alpha \subseteq R \text{ and } \beta \subseteq R$$

- The **functional dependency** $\alpha \rightarrow \beta$ **holds on** R iff for any legal relations $r(R)$, whenever any two tuples t_1 and t_2 of r agree on the attributes α , they also agree on the attributes β .

$$t_1[\alpha] = t_2[\alpha] \Rightarrow t_1[\beta] = t_2[\beta]$$

- Example:
 - A *student* table with attributes: ***ID***, ***name***, ***tot_cred***.
 - **functional dependency**: ***ID*** uniquely identifies the ***name*** attribute of student table.

$$ID \rightarrow name$$

The attributes of a table is said to be dependent on each other when an attribute of a table uniquely identifies another attribute of the same table.

Functional Dependencies

- **Superkey** K : a set of unique attributes for relation schema R if and only if $K \rightarrow R$
- K is a candidate key for R iff
 - $K \rightarrow R$, and
 - for no $\alpha \subset K, \alpha \rightarrow R$
- Example:

inst_dept (ID, name, salary, dept_name, building, budget)

dept_name $\rightarrow$ *building*



ID $\rightarrow$ *building*



dept_name $\rightarrow$ *salary*



ID	course_id	credit	instructor
210101	1	3	Park
210103	2	3	Lee
210102	3	2	Choi

- **Partial Dependency**

- An attribute in a table depends on only a **part of the PK** and not on the whole key.
- *Example*
 - *Instructor* depends on the *course*,
 - **PK: ID + course_id**
 - But the *instructor* only depends on *course_id*,
 - Hence the *course_id* has nothing to do with *ID*.

- **Solution to Partial Dependency:**
 - Remove columns *instructor* from the table and add it to the *Course* table.

course_id	title	instructor
1	Java	Park
2	C	Lee
3	Python	Choi

ID	course_id	score
210101	1	90
210103	2	92
210102	3	99

2NF

- **Normalize** the following DB.

ID	name	dept_name	tot_cred

ID	course_id	score	instructor

ID	title

2NF

- **Partial dependency:** “*instructor*” only depended on “*course_id*” and not “ID”

ID	name	dept_name	tot_cred

ID	course_id	score

course_id	title	instructor

Third Normal Form (3NF)

- **Conditions:**
 - Be in **2NF**
 - No **Transitive Dependency**.
 - An attribute is functionally dependent on another attribute which is not a part of the PK.

ID	Name	dept_name	uni_id	uni_City
----	------	-----------	--------	----------

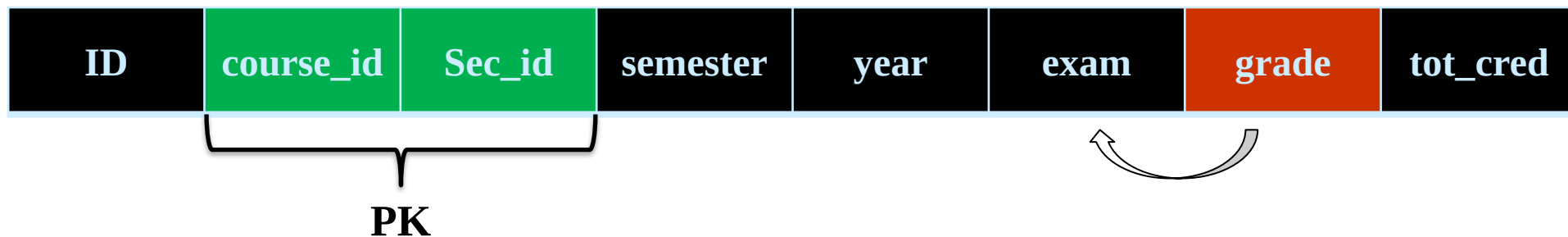
- The *Name* and *dept_name* are directly dependent on the PK, e.g. *ID*.
- while *uni_id* is not dependent.
- *uni_city* is only dependent on the *uni_id* column, and not on the PK.

3NF

- “Takes” table:

ID	course_id	sec_id	semester	year	exam	grade	tot_cred
----	-----------	--------	----------	------	------	-------	----------

- “*exam*” depends on the PK, e.g. *ID + course_id*.
- Example: For Java, there is a Practical exam but for not for Math, e.g. “*exam*” depends on “*course*”.
- “*grade*” depends on “*exam*”, which is not in a part of the key.



3NF

ID	course_id	sec_id	semester	year	exam	grade	tot_cred
----	-----------	--------	----------	------	------	-------	----------



ID	name	dept_name	tot_cred
----	------	-----------	----------

course_id	title	instructor
-----------	-------	------------

ID	course_id	grade	exam
----	-----------	-------	------

exam	score
------	-------

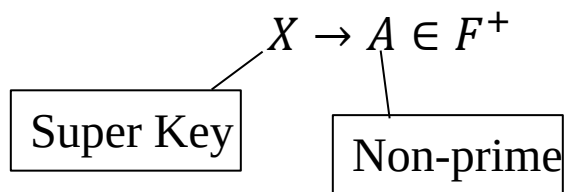
Boyce-Codd Normal Form (BCNF)

- a.k.a. **3.5 Normal Form** (3.5NF).
- A relation schema R is in BCNF with respect to a set F of functional dependencies if:
 - For all functional dependencies in F^+ of the form $\alpha \rightarrow \beta$
 - Where $\alpha \subseteq R$ and $\beta \subseteq R$, at least one of the following holds:
 - $\alpha \rightarrow \beta$ is trivial, i.e., $\beta \subseteq \alpha$
 - α is a superkey for R
- Example:
instr_dept (ID, name, salary, dept_name, building, budget)
- NOT in BCNF form!
 - $dept_name \rightarrow building, budget$ holds on *instr_dept*, but *dept_name* is not a super key!

BCNF

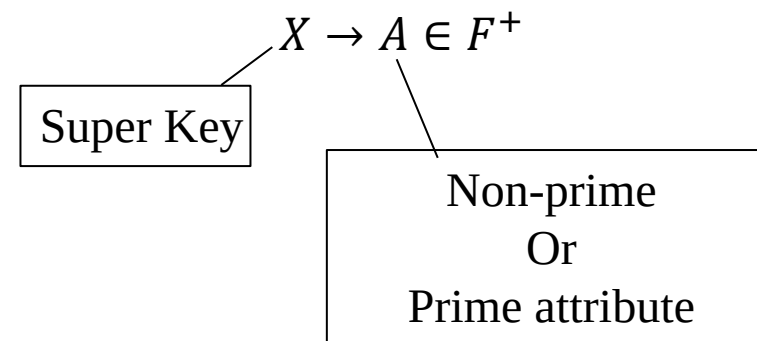
- Conditions:
 - Be in **3NF**.
 - For any dependency $A \rightarrow B$, A should be a super key.
 - e.g. A cannot be non-prime attribute if B is a prime attribute

3NF



vs.

3.5NF



Rules for BCNF

PK



ID	course_id	instructor
210101	001	Park
210101	002	Lee
210102	001	Choi

- One student can enroll for multiple courses.
 - 210101 has opted for courses 001 (Java) and 002 (Python).

BCNF

ID	course_id	instructor
210101	001	Park
210101	002	Lee
210102	001	Choi

- PK: *ID* + *course_id*,

BCNF

ID	course_id	instructor
210101	001	Park
210101	002	Lee
210102	001	Choi

$(ID, course_id) \rightarrow instructor$

Candidate Key

$instructor \rightarrow course_id$

**Non-prime
attribute**

**Prime
attribute**

BCNF

ID	course_id	instructor
210101	001	Park
210101	002	Lee
210102	001	Choi

- This relation satisfies:
 - **1NF**
 - All the values are atomic,
 - Attributes are unique
 - All the values stored in a particular column are of same domain.
 - **2NF**
 - No Partial Dependency.
 - **3NF**
 - No Transitive Dependency
 - But is NOT in **BCNF**.

BCNF

ID	Course_id	instructor
210101	001	Park
210101	002	Lee
210102	001	Choi

- PK: ID + *course_id*
 - e.g. *course_id* is a prime attribute

BCNF

- **Solution:**
 - Decompose the table into two tables

ID	course_id	instructor
210101	001	Park
210101	002	Lee
210102	001	Choi



ID	instructor_id
210101	6351
210101	6152
210102	6153

instructor_id	instructor_name	course_id
6351	Parl	001
6152	Lee	002
6153	Choi	001

Forth Normal Form (4NF)

- **Conditions:**
 - Be in **BCNF**.
 - No **Multi-valued Dependency**.
 - For a dependency $A \twoheadrightarrow B$, if for a single value of A , multiple value of B exists.
 - There should be at least 3 attributes.
 - For a relation $R(A, B, C)$, if there is a multi-valued dependency, A and B , the B and C , should be independent of each other.

4NF

ID	course_id	hobby
210101	001	Travelling
210101	002	Travelling
210102	001	Taekwondo

- Repetition in hobby attribute!
- No relationship between the *course_id* and *hobby*.

4NF

- Solution:**

ID	course_id	hobby
210101	001	Travelling
210101	002	Travelling
210102	001	Taekwondo



ID	course_id
210101	001
210101	002
210102	001

ID	hobby
210101	Travelling
210102	Taekwondo

Database Normalization Examples

- Assume a video library maintains a database of movies rented out.
- Without any normalization, all information is stored in one table as shown below.

Full name	Physical address	Movies rented	Salutation	Category
Janet jones	First street plot no. 4	Pirates of the Caribbean, clash of the titans	Ms.	Action, Action
Robert Phil	3 rd street 34	Forgetting sarah marshal, daddy's little girls	Mr.	Romance, romance
Robert phil	5 th avenue	Clash of the titans	Mr.	Action

Database Normalization Examples

- **1NF**
 - Atomicity.
 - Unique.

Full name	Physical address	Movies rented	Salutation	Category
Janet jones	First street plot no. 4	Pirates of the Caribbean	Ms.	Action
Janet jones	First street plot no. 4	clash of the titans	Ms.	Action
Robert Phil	3 rd street 34	Forgetting sarah marshal	Mr.	Romance
Robert Phil	3 rd street 34	daddy's little girls	Mr.	Romance
Robert phil	5 th avenue	Clash of the titans	Mr.	Action

Key

- **Composite Key**
 - A PK composed of multiple columns used to identify a record uniquely
- There are two people with the same name Robert Phil, but they live in different places.

Full name	Physical address	Movies rented	Salutation	Category
Robert Phil	3 rd street 34	daddy's little girls	Mr.	Romance
Robert phil	5 th avenue	Clash of the titans	Mr.	Action

Composite Key

- **2NF**

- Be in 1NF
- Single Column Primary Key

Members

Members hip_ID	Full name	Physical address	Salutation
1	Janet jones	First street plot no. 4	Ms.
2	Robert phil	3 rd street 34	Mr.
3	Robert phil	5 th avenue	Mr.

Movies

Membership_ID	Movies rented
1	Pirates of the Caribbean
1	Clash of the titans
2	Forgetting sarah marshal
2	Daddy's little girls
3	Clash of the titans

Foreign Key

- **Foreign Key**
 - *Membership_ID* is the FK

Membership_ID	Movies rented
1	Pirates of the Caribbean
1	Clash of the titans
2	Forgetting sarah marshal
2	Daddy's little girls
3	Clash of the titans

FK

Membership_ID	Movies rented
1	Pirates of the Caribbean
1	Clash of the titans
2	Forgetting sarah marshal
2	Daddy's little girls
3	Clash of the titans

PK

Members hip_ID	Full name	Physical address	Salutation
1	Janet jones	First street plot no. 4	Ms.
2	Robert phil	3 rd street 34	Mr.
3	Robert phil	5 th avenue	Mr.

Transitive Functional Dependencies

- Changing the non-key column Full Name may change Salutation:

Members hip_ID	Full name	Physical address	Salutation
1	Janet jones	First street plot no. 4	Ms.
2	Robert phil	3 rd street 34	Mr.
3	Robert phil	5 th avenue	Mr.

Changes in name may change in salutation

- **3NF**
 - Be in 2NF
 - No transitive functional dependencies

Membership_ID	Full name	Physical address	Salutation_id
1	Janet jones	First street plot no. 4	2
2	Robert phil	3 rd street 34	1
3	Robert phil	5 th avenue	1

Membership_ID	Movies rented
1	Pirates of the Caribbean
1	Clash of the titans
2	Forgetting sarah marshal
2	Daddy's little girls
3	Clash of the titans

Salutation_id	Movies rented
1	Mr.
2	Ms.
3	Mrs.
4	Dr.

- **Boyce-Codd Normal Form (BCNF)**

- Even when a database is in 3rd Normal Form, still there would be anomalies resulted if it has more than one **Candidate** Key.
- Sometimes BCNF is also referred as 3.5 Normal Form.

- **4NF (Fourth Normal Form) Rules**

- If no database table instance contains two or more, independent and multivalued data describing the relevant entity, then it is in 4th Normal Form.

- **5NF (Fifth Normal Form) Rules**

- A table is in 5th Normal Form only if it is in 4NF and it cannot be decomposed into any number of smaller tables without loss of data.

- **6NF (Sixth Normal Form) Proposed**

- 6th Normal Form is not standardized, yet however, it is being discussed by database experts for some time.

Summary

Form	Acronym	Challenge
First Normal Form	1NF	Atomicity, unique attribute name, same domain
Second Normal Form	2NF	Partial Dependency
Third Normal Form	3NF	Transitive Dependency
Boyce-Codd Normal Form	BCNF	Non-prime Dependency
Forth Normal Form	4NF	Multi-valued Dependency
Fifth Normal Form	5NF	Decomposable
Sixth Normal Form	6NF	TBA



Md. Jalil Piran, Ph.D.

Professor of Artificial Intelligence,
IEEE Senior Member,

Editor of IEEE Transactions on Intelligent Transportation Systems,

Editor of Journal of Engineering Applications of Artificial Intelligence,

Vice-Chair of the IEEE Consumer Technology Society on ML, DL, and AI in Consumer Electronics.

piran.mj2@gmail.com